

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	AI-Driven Self-Healing Agent for Microservice Vulnerabilities
Student ID:	00053080
Student name:	Hadrian Ging Hin Chong
Student email	hadrianghinghin.chong2@mail.dcu.ie
Student 2 ID:	21360701
Student 2 name:	Eamon Cullimore
Student 2 email	eamon.cullimore2@mail.dcu.ie
Chosen major:	MCM Artificial Intelligence
Supervisor	Yalemisew Abgaz
Date of Submission	21/11/2025

Section 2: About your Practicum

What is the topic of your proposed practicum?

The proposed practicum will be focusing on the development of an AI-driven self-healing agent for microservices, using the Train-Ticket benchmark system and DeepTraLog dataset. The project aims to design and implement supervised anomaly detection models that identify faults in service traces and logs. Detected issues which include asynchronous, multi-instance, configuration, and code-level faults are automatically diagnosed and remediated by the agent using rule-based or adaptive policies. Integration with orchestration APIs will enable real-time recovery actions such as restarts and rollbacks. The work will evaluate the improvement in service resilience, fault recovery speed, and downtime reduction compared to manual incident response and traditional monitoring.

Please provide details of the papers you have read on this topic

1. *QMConn: A Self Healing Controller for Microservices Using Q-Learning and Markov Decision Processes:*

- Adaptive RL policy for configuration repair.
- Deployed on Kubernetes with induced configuration errors: Incorrect Configuration (INC), Weak Access Control (WAC), etc.
- MTTR: 13 min (single) → 23.75 min (multiple).
- The system tolerated cascading failures.
- Optimized recovery cost-performance.

2. Self-Healing Microservices Architecture: Automated Monitoring and Rollback in Kubernetes and GCP:

- Telemetry-driven anomaly detection using pod health & latency.
- Automated rollback triggered by Kubernetes events.
- Adaptive scaling for resource efficiency.
- 92% anomaly detection accuracy, compared to around 70% with traditional methods
- Mean time to defect failures (MTTD) dropped by 50%,
- Mean time to recovery (MTTR) improved by 30–40%.
- Recovery: 12–15s (vs 3 min manual)

3. An Intelligent Fault Self Healing Mechanism for Cloud AI Systems via Integration of Large Language Models and Deep Reinforcement Learning

- LLM encoder (7B) interprets fault semantics.
- Proximal policy optimization (PPO) based deep RL optimizes recovery actions.
- Memory-guided meta-control for adaptation.
- Fault detection accuracy: 0.89–0.92.
- Recovery time reduced from 100 s to 85 s.

4. A New Era in Software Security: Towards Self-Healing Software via Large Language Models and Formal Verification.

- Developed a new framework (ESBMC-AI) that used formal verification to detect, identify and resolve faults in code.
- Focused heavily on training a model on arithmetic and how to resolve errors in C code.
- Tested on Kubernetes environment with over 50,000 C programs

5. Phoenix Stack: A Self-Healing Microservice Architecture for Real-Time Web Applications:

- Developed and built a web-service framework that operated through multiple agents throughout a microservice architecture system.
- The agents each detect, identify and resolve issues independently and communicate with each other throughout the process.
- Utilized several tools and environments such as: Kubernetes, AWS Cloud, Python, Go and Node.js.

How does your proposal relate to existing work on this topic described in these papers?

Our practicum will be building directly on key advances highlighted in the recent literature review on AI-driven self healing for microservices. Like QMConn and Phoenix Stack, we will apply reinforcement learning and supervised anomaly detection to automate fault recovery in Kubernetes orchestrated microservice environment. Besides, our approach integrates proven methods such as telemetry-based anomaly detection mentioned in Paper 2, adaptive policy mapping, and automated remediation workflows, as demonstrated by high-accuracy, fast recovery solutions in the reviewed students.

Furthermore, unlike papers that focus exclusively on configuration or code faults such as in Paper 1 and Paper 4, our work will address multiple real-world fault scenarios (async, multi-instance, config, and monolithic) within the Train-Ticket benchmark, using both trace and log analytics.

Besides, Paper 3 and Paper 5 had multi-agent and LLM-guided control. Our agent will benchmark different detection models and remediation strategies, seeking to match or exceed state-of-the-art results in recovery time, reliability, and operational efficiency.

What is the research question that you will attempt to answer?

How can AI-driven self-healing frameworks autonomously detect, diagnose, and remediate common faults in microservices? This practice focuses on evaluating whether trained AI-agents integrated into a real-world microservice can outperform current manual approaches such as human-administered monitoring, log inspection, and incident response in terms of fault recovery speed, downtime reduction, and operational reliability. The core question would be whether AI-powered automation can deliver measurable, practical improvements in resilience and efficiency over manual, rule-based, or ad-hoc management techniques used by operators and developers.

How will you explore these questions?

What software and programming environment will you use?

The practicum will use Python for AI model development and integration, utilizing libraries such as scikit-0., TensorFlow or PyTorch for supervised anomaly detection. Distributed tracing and log analytics will utilize tools like ElasticSearch, in conjunction with the DeepTraLog dataset. The Train-Ticket microservice will be deployed on Kubernetes or Docker Compose for orchestration and automation of self-healing actions. Orchestration APIs will be accessed using the Kubernetes Python client. Additional monitoring may employ Prometheus for real-time metric visualization.

What coding/development will you do?

First and foremost, we will preprocess the DeepTraLog dataset and implement supervised machine learning models to detect anomalies and classify fault types in microservice traces and logs. Next we will develop a self-healing agent module that integrates the trained AI model, continuously ingests live trace or log data from ElasticSearch, and triggers automated remediation actions such as service restarts, configuration rollbacks, and replica synchronization by interfacing with the Kubernetes orchestration API. Furthermore, we will write code for real-time monitoring, result reporting, and validation experiments to compare automated healing versus manual remediation. All code will be packaged for integration with the Train-Ticket microservice system.

What data will be used for your investigations?

Provided by **DeepTraLog** where the dataset is generated from [TrainTicket](#), it has **132,485** traces and **7,705,050** log messages. Among the traces **23,334 (17.6%)** are anomalous.

Is this data currently available, if not, where will it come from?

Data is available at (<https://github.com/FudanSELab/DeepTraLog>) while the source code of the dataset is available at (<https://github.com/FudanSELab/train-ticket>)

What experiments do you expect to run?

We will be running experiments to compare the effectiveness of AI-driven self-healing against manual remediation in managing faults within the Train-Ticket microservice system. This will include injecting fault scenarios such as asynchronous, multi-instance, configuration and monolithic errors and measure the time to detect, diagnose and recover using the self-healing agent versus traditional human intervention. Besides, we will quantify operational metrics such as downtime duration, error rate, and recovery speed. Experiments will also benchmark different anomaly detection models to identify the best approach for real time fault detection. The performance, resource usage, and impact on service reliability will be analyzed and reported.

What output do you expect to gather?

We will be gathering quantitative data and visual reports comparing the effectiveness of AI-driven self-healing versus manual remediation in microservices. Output will include the accuracy and speed of anomaly detection, fault diagnosis logs, time-to-recover and downtime metrics, and before-and-after service availability statistics for each fault type. Additional results will consist of automated action logs such as restarts, rollbacks and others, performance benchmark for different anomaly

detection models and summary dashboards highlighting overall system resilience improvement. All of the analysis will be documented to demonstrate how autonomous agents optimize operational efficiency and reliability in real-world microservice deployments.

How will the results be evaluated?

Results will be evaluated using both quantitative and qualitative metrics. The key criteria include fault detection accuracy, reduction in downtime, speed of automated recovery, and overall service availability. The comparative experiments will measure how quickly and effectively the self-healing agent resolves faults versus manual remediation, using time-to-repair and system reliability benchmarks.

On the other hand, statistical analysis will be carried out on operational logs, anomaly detection performance, and service health data. The experiments will be validated through repeatable fault-injection scenarios, while dashboards and visualizations will support demonstration of improvements in operational efficiency and resilience.

References

- [1] A. Samir, H. Dagenborg, and D. Johansen, "QMConn: A Self-Healing Controller for Microservices Using Q-Learning and Markov Decision Processes," in 2024 IEEE/ACM 17th International Conference on Utility and Cloud Computing (UCC), Dec. 2024, pp. 389–398. doi: 10.1109/UCC63386.2024.00060.
- [2] S. K. Chintakindhi, "Self-Healing Microservices Architecture: Automated Monitoring and Rollback in Kubernetes and GCP," *Int. J. Emerg. Trends Comput. Sci. Inf. Technol.*, pp. 9–23, Sept. 2025, doi: 10.63282/WCAI25-102.
- [3] Z. Yang, Y. Jin, J. Liu, and X. Xu, "An Intelligent Fault Self-Healing Mechanism for Cloud AI Systems via Integration of Large Language Models and Deep Reinforcement Learning," June 09, 2025, arXiv: arXiv:2506.07411. doi: 10.48550/arXiv.2506.07411.
- [4] N. Tihanyi, Y. Charalambous, R. Jain, M. A. Ferrag, and L. C. Cordeiro, "A New Era in Software Security: Towards Self-Healing Software via Large Language Models and Formal Verification," pp. 136–147, Apr. 2025, doi: <https://doi.org/10.1109/ast66626.2025.00020>.
- [5] Bekir Tolga Tutuncuoglu, "Phoenix Stack: A Self-Healing Microservice Architecture for Real-Time Web Applications," Jan. 2025, doi: <https://doi.org/10.2139/ssrn.5343101>.